

Project Gutenberg language diversity over time

Dirk van Wezel
S4910583

Abstract

1 Introduction

This study analyses the linguistic diversity in Project Gutenberg's digital library collection over time, examining changes in the additions for each language in the Gutenberg Collection per year. The research aims to understand how increased internet access in non-English speaking countries may have influenced digital preservation efforts. It does this by analysing the works added year over year to the Gutenberg collection, showing the changes in addition rates of differing languages. The hypothesis is that over time, the proportion of English books will decrease compared to non-English books, reflecting an increase in internet availability and interest in preservation of historical books, and a decrease in available public domain books written in English. On the frequently asked questions pages on the [gutenberg.org](https://www.gutenberg.org) website they explain how books are added, namely by volunteers. This makes it difficult for Project Gutenberg to directly change which books are added, as they can't dictate what these volunteers add.

2 Related Work

Several studies have examined linguistic diversity in digital libraries and Project Gutenberg specifically:

1. Whitt (2018) explored multilingualism on the web and Project Gutenberg's early efforts at linguistic diversification, providing important historical context for understanding the evolution of digital preservation efforts.
2. Gerlach and Font-Clos (2020) analysed a curated Project Gutenberg database, revealing a strong bias toward Western languages, with English comprising 81% of all books. Their work established baseline metrics for understanding linguistic representation in digital libraries.

3 Data

Pre-processing The data pre-processing was done by downloading the metadata of all Project Gutenberg books from the [gutenberg.org](https://www.gutenberg.org) website on 06-01-2025. This data can be downloaded in the form of a CSV file, which contains the relevant fields for this article: the date added and the language. Books that included multiple languages, either due to being a translation that includes the original text or being a translation dictionary, are counted for both languages. The

python script used for extracting the data will be included with the paper. Due to not being completed, and thus giving incomplete information, the year 2025 was excluded. As only the language codes were given with the metadata, language codes were manually linked to their respective language. This resulted in a matrix for book additions per year for every language, from the years 1971 to 2024, with 70 languages. Many of these languages have very few books however. For example, of the 70 languages 46 have less than 10 total books. This makes any analysis of including these languages less likely to be significant.

4 Predicted Results

We expect that the data will show an increase in the proportion of non-English books in Project Gutenberg's collection over time. This is based on the idea that increased internet access in non-English speaking countries leads to greater easier access and participation with Project Gutenberg. Additionally, we anticipate that the diversity of languages represented will have broadened, reflecting a more globalized effort in digital preservation.

Discussion

5 Conclusion

References

Gerlach, Martin and Francesc Font-Clos (Jan. 2020). "A Standardized Project Gutenberg Corpus for Statistical Analysis of Natural Language and Quantitative Linguistics". In: *Entropy* 22.1. Number: 1 Publisher: Multidisciplinary Digital Publishing Institute, p. 126. ISSN: 1099-4300. DOI: 10.3390/e22010126. URL: <https://www.mdpi.com/1099-4300/22/1/126> (visited on 12/12/2024).

Whitt, Richard J. (2018). *Diachronic Corpora, Genre, and Language Change*. Amsterdam/Philadelphia, NETHERLANDS, THE: John Benjamins Publishing Company. ISBN: 978-90-272-6350-6. URL: <http://ebookcentral.proquest.com/lib/rug/detail.action?docID=6733930> (visited on 12/05/2024).

6 Appendix

6.1 Appendix A

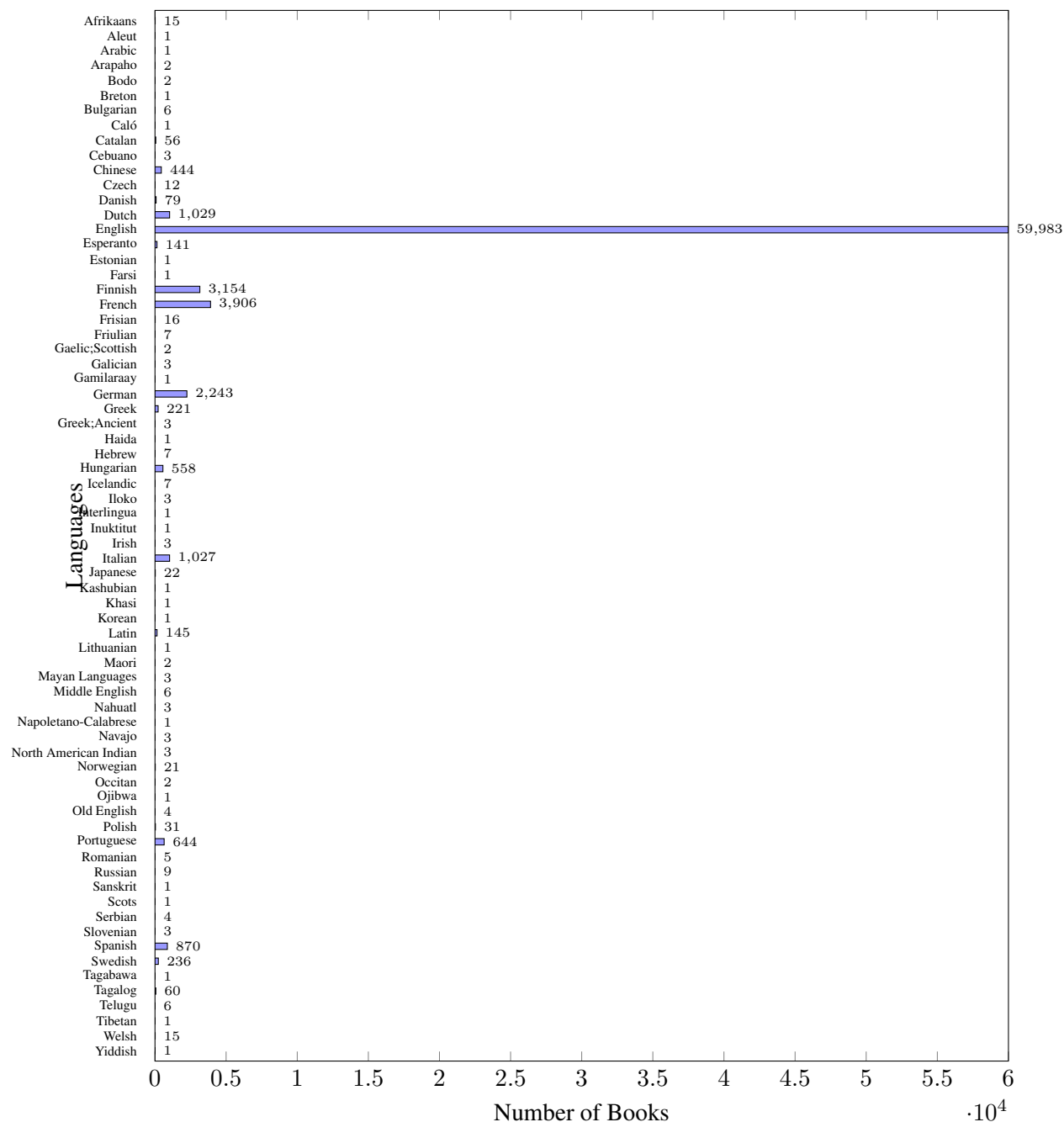


Figure 1: Total books per language in the Gutenberg Corpus

Language	1971	1972	1973	1975	1976	1977	1978	1979	1980	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
English	1	2	1	1	1	1	1	1	1	5	13	36	58	102	334	315	348	339	311	313	376	1373	6363	4273	2887	3203	2798	2507	3382	3007	2230	2474	2194	2237	1865	1770	2013	2280	2007	1983	2207	1573	
French	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
German	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Spanish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Italian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Portuguese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Hungarian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Chinese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Swedish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Greek	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Hebrew	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Japanese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Norwegian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Dutch	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Finnish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Polish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Czech	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Russian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ukrainian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Belarusian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Georgian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Armenian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Abkhazian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ossetian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Kabardian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Chechen	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Dagestanian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	133	2207	1573
Ingush	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	1	39	328	263	319</																	

Table 1: The number of books added to Project Gutenberg per year per language